

Relazioni di ricorrenza

Rel. di ric.	$\mathcal{O}$ -grande	Esempi
$T(n) = T(n-1) + 1$	$\mathcal{O}(n)$	maximo/minimo
$T(n) = T(n-1) + n$	$\mathcal{O}(n^2)$	insert sort
$T(n) = 2T(n-1) + 1$	$\mathcal{O}(2^n)$	Hanoi
$T(n) = T(\frac{n}{2}) + 1$	$\mathcal{O}(\log n)$	ric. dic.
$T(n) = T(\frac{n}{2}) + n$	$\mathcal{O}(n)$	
$T(n) = 2T(\frac{n}{2}) + 1$	$\mathcal{O}(n)$	
$T(n) = 2T(\frac{n}{2}) + n$	$\mathcal{O}(n \log n)$	merge sort

$$T(n) = T(n-1) + 1$$

$$= T((n-1)-1) + 1 + 1 = T(n-2) + 2$$

$$= T(n-k) + k \quad k \leq n$$

$$= T(0) + n$$

$$= O(n)$$

$$\text{se } T(0) = O(1)$$

$$T(n) = T(n-1) + n$$

$$T(n) = T(n-1) + n$$

$$= T((n-1)-1) + n-1 + n$$

$$= T(n-2) + n-1 + n$$

$$= T((n-2)-1) + n-2 + n-1 + n$$

$$= T(n-3) + n-2 + n-1 + n$$

$$\underbrace{\quad}_{\sum_{i=0}^{3-1} (n-i)}$$

$$= T(n-k) + \sum_{i=0}^{k-1} (n-i) \quad \text{per } k \leq n$$

$$= T(0) + \sum_{i=0}^{n-1} (n-i) = 1 + \sum_{i=1}^n i = 1 + \frac{n(n+1)}{2} \in O(n^2)$$

$$\sum_{i=0}^{n-1} (n-i) = n + (n-1) + (n-2) + \dots + \underbrace{(n-(n-1))}_1$$

$$= 1 + 2 + \dots + n$$

$$= \sum_{i=1}^n i$$

$$T(n) = 2T(n-1) + 1$$

$$T(n) = 2T(n-1) + 1$$

$$= 2[2T(n-2) + 1] + 1$$

$$= 4T(n-2) + 2 + 1$$

$$= 4[2T(n-3) + 1] + 2 + 1$$

$$= 8T(n-3) + 4 + 2 + 1$$

$$= 2^3 T(n-3) + \underbrace{2^2 + 2^1 + 2^0}_{\sum_{i=0}^{3-1} 2^i}$$

$$= 2^k T(n-k) + \sum_{i=0}^{k-1} 2^i$$

$$k \leq n$$

$$= 2^n T(0) + \sum_{i=0}^{n-1} 2^i$$

$$= 2^n + 2^n - 1 = O(2^n)$$

$$\sum_{i=0}^n x^i = \frac{x^{n+1} - 1}{x - 1}$$

where $x \neq 1$

$$T(n) = T\left(\frac{n}{2}\right) + 1$$

$$T(n) = T\left(\frac{n}{2}\right) + 1$$

$$= \left[T\left(\frac{\frac{n}{2}}{2}\right) + 1 \right] + 1$$

$$= T\left(\frac{n}{4}\right) + 1 + 1 = T\left(\frac{n}{4}\right) + 2$$

$$= T\left(\frac{n}{8}\right) + 1 + 2 = T\left(\frac{n}{8}\right) + 3$$

$$= T\left(\frac{n}{2^k}\right) + k \quad k \leq \log_2 n$$

$$= T(1) + \log_2 n = O(\log n)$$

$$T(n) = T\left(\frac{n}{2}\right) + n$$

$$= T\left(\frac{n}{4}\right) + \frac{n}{2} + n$$

$$= T\left(\frac{n}{8}\right) + \frac{n}{4} + \frac{n}{2} + n$$

$$= T\left(\frac{n}{2^3}\right) + \underbrace{\frac{n}{2^2} + \frac{n}{2^1} + \frac{n}{2^0}}$$

$$\sum_{i=0}^{3-1} \frac{n}{2^i}$$

$$= T\left(\frac{n}{2^k}\right) + \sum_{i=0}^{k-1} \frac{n}{2^i}$$

$$k \leq \log_2 n$$

$$a^{(\log a(b))} = b$$

$$T(n) = T\left(\frac{n}{2}\right) + n$$

$$= T\left(\frac{n}{2^{\log_2 n}}\right) + \dots +$$

$$= T(1) + \sum_{i=0}^{\log_2 n - 1} \frac{n}{2^i}$$

$$= 1 + n \sum_{i=0}^{\log_2 n - 1} \left(\frac{1}{2}\right)^i$$

$$= 1 + n \frac{\left(\frac{1}{2}\right)^{\log_2 n} - 1}{\frac{1}{2} - 1}$$

$$= 1 + n \frac{1 - 1/n}{1/2}$$

$$= 1 + n \left(2 - \frac{2}{n}\right) = 1 + 2n - 2$$

$$T(n) = 2T\left(\frac{n}{2}\right) + 1$$

$$T(n) = 2T\left(\frac{n}{2}\right) + 1$$

$$= 2\left[2T\left(\frac{n}{4}\right) + 1\right] + 1$$

$$= 4T\left(\frac{n}{4}\right) + 2 + 1$$

$$= 4\left[2T\left(\frac{n}{8}\right) + 1\right] + 2 + 1$$

$$= 8T\left(\frac{n}{8}\right) + 4 + 2 + 1$$

$$= 2^k T\left(\frac{n}{2^k}\right) + \sum_{i=0}^{k-1} 2^i$$

$$k \leq \log_2 n$$

$$= n + \sum_{i=0}^{\log_2 n - 1} 2^i = n + 2^{\log_2 n} - 1 = 2n - 1 \in O(n)$$

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$

$$= 2\left[2T\left(\frac{n}{4}\right) + \frac{n}{2}\right] + n$$

$$= 4T\left(\frac{n}{4}\right) + 2n$$

$$= 4\left[2T\left(\frac{n}{8}\right) + \frac{n}{4}\right] + 2n$$

$$= 8T\left(\frac{n}{8}\right) + 3n$$

$$= 2^k T\left(\frac{n}{2^k}\right) + kn$$

$$k \leq \log_2 n$$

$$= n + n \log_2 n \in O(n \log n)$$

$$\frac{n}{2^k} \geq 1$$

$$n \geq 2^k$$

$$\log_2 n \geq \log_2 2^k = k$$